E-ink DAP player board, Phase 2

Generated by tools/make_schematic.py from the plan in docs/pcb_plan.md. Change the script, not the sheets.

Sheets connect through global labels with the same name; supply nets use power symbols.

Footprints are the chosen parts' own (see docs/pcb_plan.md); both headphone jacks sit on the back.

Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Display

File: display.kicad_sch

USB and storage

File: usb_storage.kicad_sch

Audio

File: audio.kicad_sch

Sheet: /
File: dap.kicad_sch

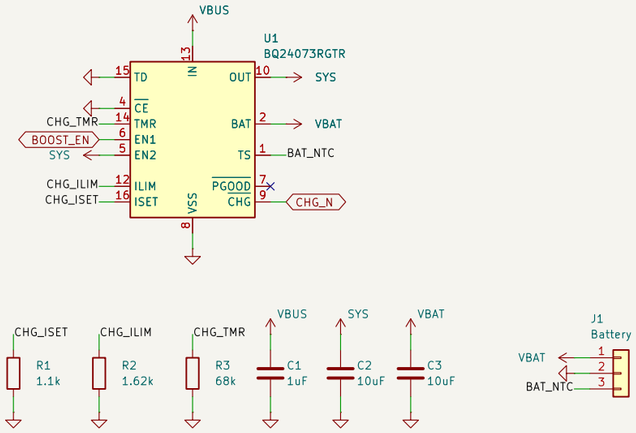
Title:

Size: A3 | Date: | **Rev:**

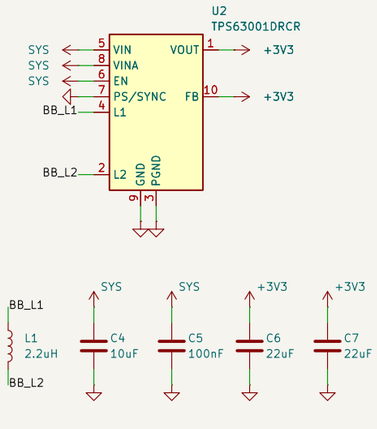
KiCad E.D.A. 10.0.7 | Id: 1/6



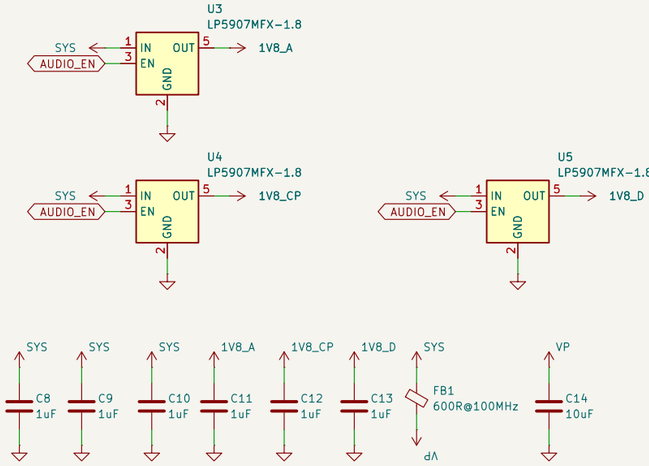
Charger and power path: BQ24073, input limit 1.0 A, charge 0.8 A, OUT (SYS) held at 4.4 V on USB



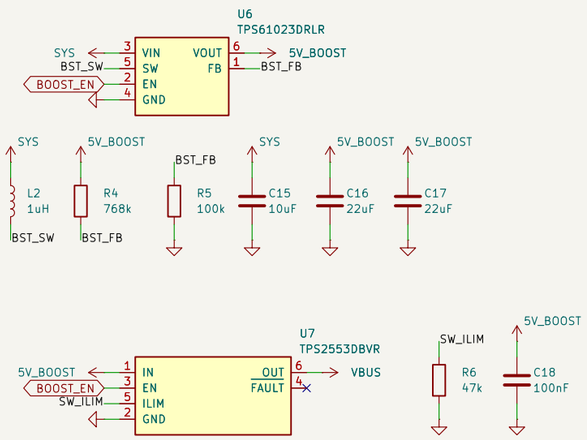
3.3 V buck–boost: TPS63001, power-save mode on; 2.2 uH as in TI's circuit (1.6 A)



Audio rails: three LP5907 1.8 V LDOs (VA, VCP, VD+VL) and the filtered VP rail, off when AUDIO_EN is low

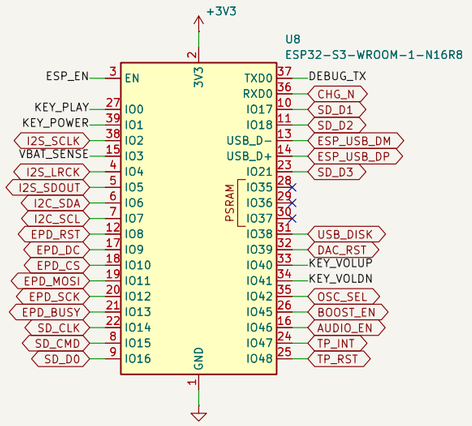


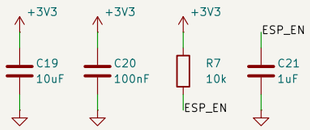
Optional 5 V out for a USB DAC: TPS61023 boost (5.1 V) + TPS2553 limit ~500 mA (47k); leave unassembled if not needed



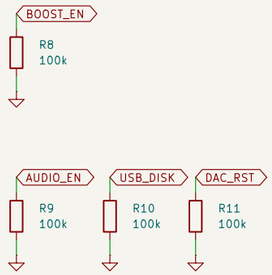
Sheet: /Power/		
File: power.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 2/6

ESP32-S3-WROOM-1-N16R8 (pin map: docs/pcb_plan.md). IO35-37 belong to the octal PSRAM

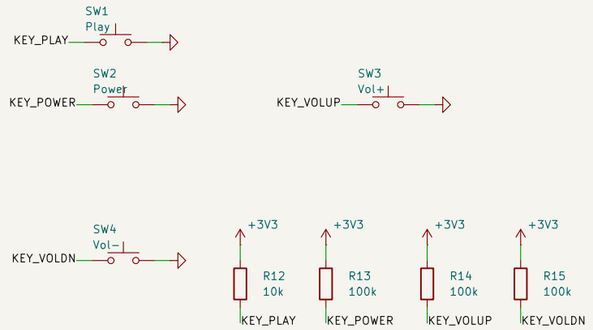




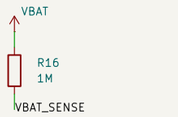
Boot straps and defaults: IO45 and IO46 must be low at boot; USB_DISK, DAC_RST, AUDIO_EN, BOOST_EN default low

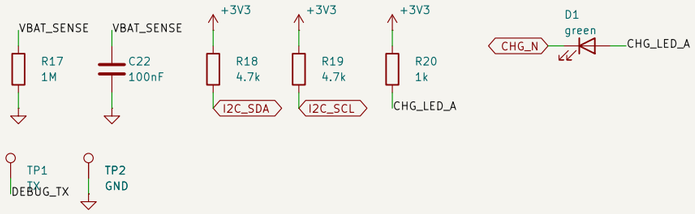


Side keys (Alps SKRTLAE010): pressed = low. Play is on IO0, so holding it at power-on starts download mode



Battery voltage (1M/1M divider, about 2 uA), I2C pull-ups (3.3 V side), charge LED on CHG_N, debug pads





Reset pads on the back, beside the battery pads: bridge them for a moment (tweezers) to restart the ESP32

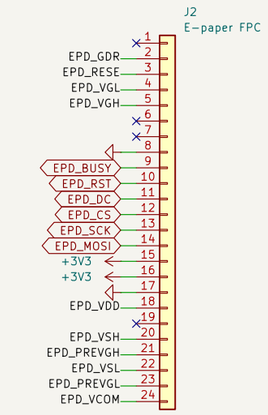


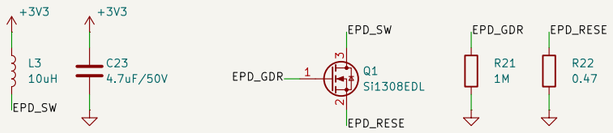
Sheet: /MCU/
File: mcu.kicad_sch

Title:

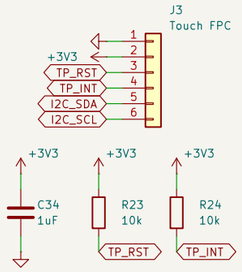
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/6

E-paper GDEY037T03 (UC8253), 24-pin FPC: the reference circuit in the GDEY037T03-T02 spec (p. 35), 0.47 R sense for UC driver ICs. BS1 low = 4-wire SPI

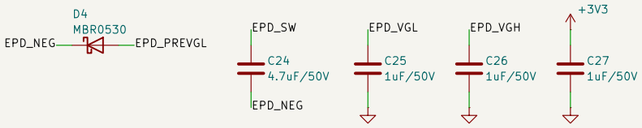


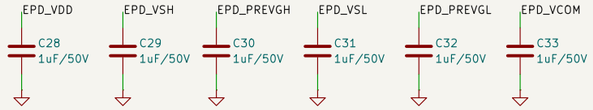


Touch FT6336U, 6-pin FPC (Good Display TP037-S02: 1 GND, 2 VDD, 3 RST, 4 INT, 5 SDA, 6 SCL), I2C address 0x38





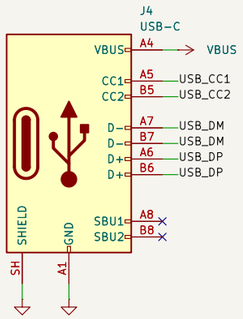


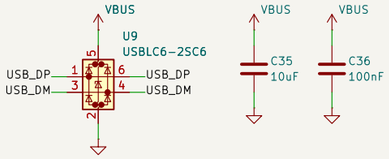


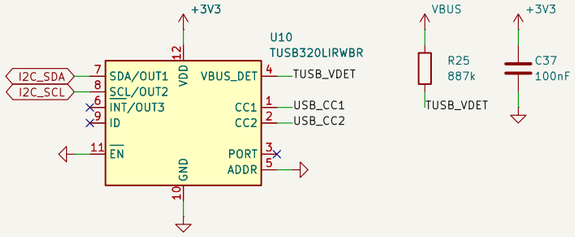
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Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 4/6



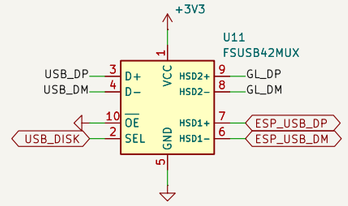
USB-C (16-pin) with ESD protection; CC handled by TUSB320L (dual-role, I2C 0x47)





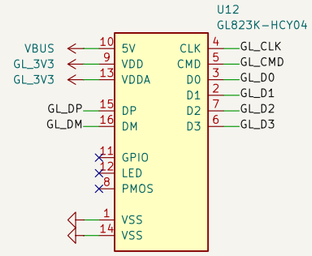


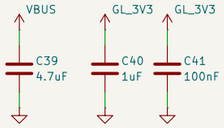
USB switch FSUSB42: USB_DISK low = ESP32, high = card reader



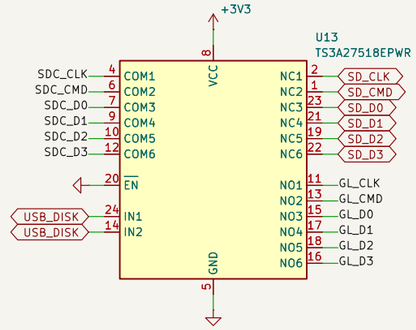


Card reader GL823K: runs from VBUS (only when a PC is plugged in), no crystal needed



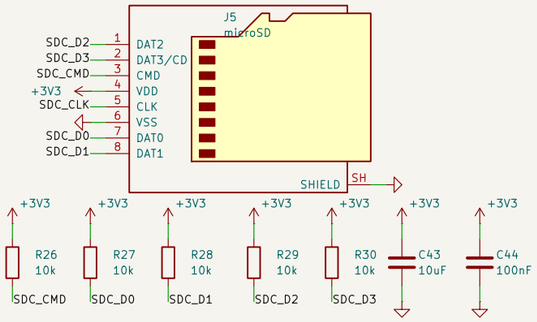


microSD switch TS3A27518E: USB_DISK low = ESP32 (NC side), high = GL823K (NO side)



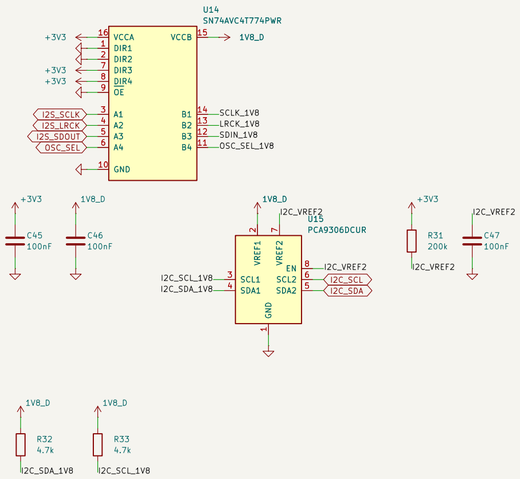


microSD socket (push-push). 10k pull-ups on CMD and DAT0-3, card powered from +3V3

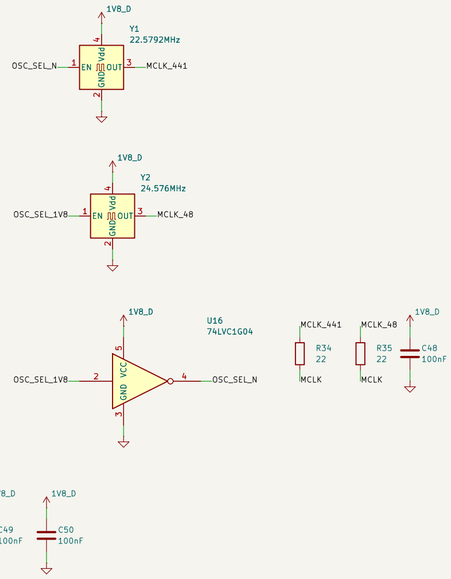


Sheet: /USB and storage/		
File: usb_storage.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 5/6

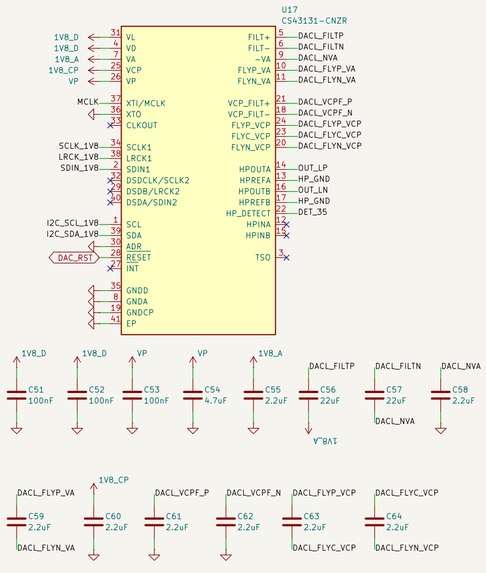
Level shifting: SN74AVC4T774 (A = 3.3 V ESP32 side, B = 1.8 V DAC side); PCA9306 joins the I2C buses



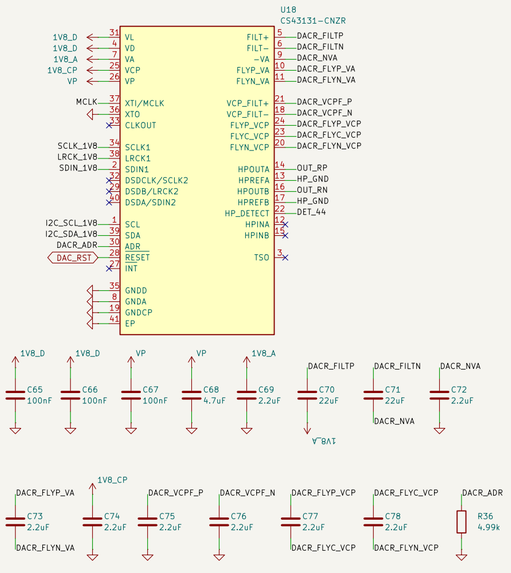
Audio clocks: one oscillator at a time, picked by OSC_SEL (low 22.5792 MHz for 44.1k music, high 24.576 MHz for 48k)



CS43131 L (I2S clock master, mono mode): HPOUTA = OUT_LP, HPOUTB = OUT_LN. Caps per Cirrus's typical connection diagram; XTO to ground (external clock)

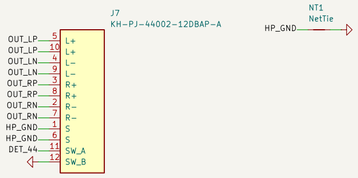


CS43131 R (I2S clock slave, mono mode): HPOUTA = OUT_RP, HPOUTB = OUT_RN. Caps per Cirrus's typical connection diagram; XTO to ground (external clock)



Headphone jacks, both on the back of the board: they share the outputs; 3.5 mm takes L+ and R+ only. HP_GND joins GND at the jacks





Sheet: /Audio/		
File: audio.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 6/6